



Shared Mental Models in Generative AI-Based Multi-Agent Systems: A Simulation Study Grounded in Human Team Cognition Theory

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I. Introduction

Collective decision making in distributed tasks depends on whether team members disclose and integrate knowledge distributed across the team. This challenge is central to complex knowledge work. LLM-based multi-agent systems (MAS) offer a computational approach to such tasks by giving agents distinct roles and information profiles. However, the mere presence of multiple agents does not guarantee effective and efficient collaboration. Agents must disclose privately held knowledge, retain relevant contributions, and integrate them into a coherent representation of the evolving task. Human team research explains such coordination through shared mental models (SMMs), which provide members with sufficiently compatible understandings of the task, the team, and their interaction. Translating this idea into a computational design mechanism, this study investigates whether a SMM-inspired structured memory improves the efficiency and effectiveness of LLM-based MAS, and how its effects depend on the amount of prior discussion history available to agents.

II. Theoretical Background & Hypotheses

Research commonly explains team performance through the input-process-outcome framework. Inputs comprise the resources and conditions available to a team, processes describe how members transform inputs into outputs, and outcomes capture the results of their collective work. One central input factor is the SMM. They provide team members with a sufficiently compatible understanding of the task, the team, and the current situation. This common cognitive basis helps members anticipate what information others need, connect individual contributions to the collective task, and coordinate their actions. Through the I-P-O framework SMM supports two central team processes, communication and cooperation, which in turn shape team performance.

LLM-based MAS face an analogous coordination problem when specialised agents must integrate distributed information into a joint decision. To overcome this problem, this study translates the mechanism of SMM into an external structured memory mechanism. It records task-relevant facts, candidate evaluations, individual positions, the emerging group view, and unresolved questions and reintroduces this representation into the agents' context before subsequent contributions. Its effects may depend on how much of the prior discussion history remains accessible alongside it. This amount is termed discussion-context transparency. Structured memory and discussion-context transparency together define the system's context configuration.

Accordingly, the structured memory is expected to improve communication quality (H1a) and cooperation quality (H1b). Within the structured memory conditions, moderate discussion context transparency is expected to outperform low transparency because agents can better trace and ground previous contributions (H2a, H2b). Moderate transparency is also expected to outperform high transparency because excessive context may reduce the salience of decision-relevant information (H3a, H3b). Finally, higher communication and cooperation quality are expected to improve both efficiency and effectiveness (H4a-H5b).

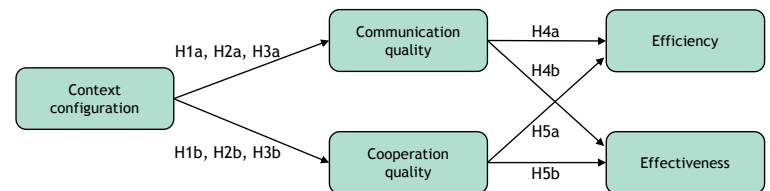


Fig. 2: Conceptual research model and hypothesized relationships

III. Methodology

The study is based on 800 independent simulations of a hidden-profile personnel-selection task. Three LLM-based agents acted as an airline hiring panel and selected one of four candidates for a long-distance pilot position. According to the hidden profile paradigm, all agents received shared task materials, while also holding private candidate information. Candidate C was the superior choice in the complete information set and could only be identified if agents disclosed and integrated their private facts. The study compared a baseline without a structured memory mechanism with three structured memory conditions that differed in discussion-context transparency: low (only the immediately preceding contribution), moderate (full public discussion history), and high (full public history plus prior reasoning traces). Communication quality, cooperation quality, weighted token usage as proxy for efficiency, and correct selection of Candidate C as proxy for effectiveness were used as outcome measures.

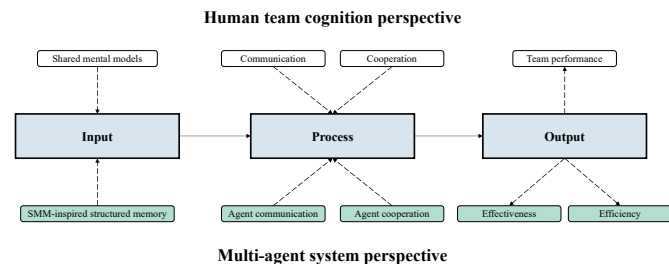


Fig. 1: Mapping the human team cognition perspective onto LLM-based MAS

IV. Results & Discussion

Three takeaways stand out. First, the SMM-inspired structured memory supports better collective decisions, although this improvement came with greater computational effort and depended on the surrounding context. The strongest outcomes emerged under low discussion-context transparency, where the structured representation was accompanied by only limited raw discussion history. Second, more context is not necessarily better. High transparency increased token usage and was associated with lower decision effectiveness, suggesting that extensive discussion history can reduce the salience of relevant information. Third, communication and cooperation play distinct roles. Communication made hidden information available and was associated with more correct decisions, but also with greater computational effort. Cooperation, by contrast, was associated with lower token usage, suggesting that references to others' contributions may help agents work more efficiently with already available information. Overall, LLM-based teams benefit less from unrestricted access to information than from a compact, structured representation of what the group knows, believes, and still needs to resolve. Context management is therefore a central design problem for LLM-based multi-agent systems: effective collaboration depends not on providing all available information, but on presenting the most relevant information in a usable form.

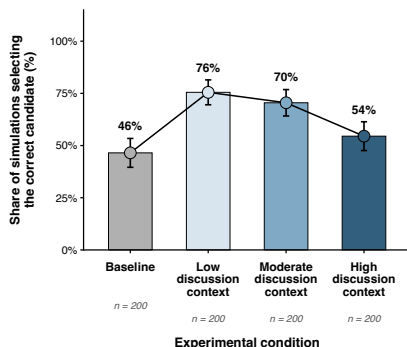


Fig. 3: Correct-decision rates across experimental conditions

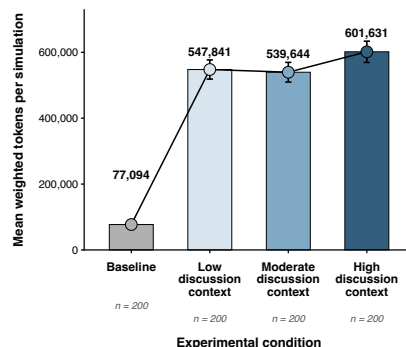


Fig. 4: Weighted token usage across experimental conditions

Model	(1)	(2)	(3)	(4)	(5)	(6)
Hypothesis	H1a	H1b	H2a/H3a	H2b/H3b	H4a/H5a	H4b/H5b
Estimator	OLS	OLS	OLS	OLS	OLS	Logit
Dependent Variable	Communication	Cooperation	Communication	Cooperation	Efficiency	Effectiveness
Intercept	0.167*** (0.006)	0.478*** (0.014)	0.220*** (0.006)	0.462*** (0.015)	0.438*** (0.024)	-0.831* (0.325)
Baseline					-0.428*** (0.015)	-0.747*** (0.222)
Low transparency	0.072*** (0.010)	0.139*** (0.020)	0.019* (0.010)	0.155*** (0.021)	0.008 (0.020)	0.064 (0.238)
Moderate transparency	0.055*** (0.009)	-0.016 (0.021)				
High transparency	0.027** (0.009)	-0.046* (0.023)	-0.026** (0.009)	-0.030 (0.024)	0.076*** (0.022)	-0.539* (0.220)
Communication					0.631*** (0.066)	6.503*** (1.017)
Cooperation					-0.081** (0.027)	0.720- (0.368)
Observations	800	800	600	600	800	800
R ²	0.082	0.096	0.040	0.115	0.603	
Adjusted R ²	0.078	0.092	0.036	0.112	0.601	

Note: HC3 standard errors in parentheses. * p < .10, * p < .05, ** p < .01, *** p < .001

Tab. 1: Regression results

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